

2D CPU/GPU Die Thermal Simulation Using Finite Difference Methods

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Abstract. This report presents a 2D transient and steady-state heat transfer simulation of a multi-layer CPU/GPU chip package using finite difference methods (FDM). Three numerical solvers are implemented in MATLAB: an explicit forward-time central-space (FTCS) scheme, an implicit backward-time central-space (BTCS) scheme solved with Gauss–Seidel iteration, and a direct steady-state Gauss–Seidel solver. The domain models three physical layers—silicon die, thermal interface material (TIM), and copper heat spreader—with three localized volumetric heat sources at $\dot{q}''' = 5 \times 10^8 \text{ W/m}^3$ representing CPU cores. Convective cooling ($h = 2,000 \text{ W/m}^2 \text{ K}$) is applied at the top and lateral boundaries with an adiabatic bottom. Peak junction temperatures, layer-to-layer thermal gradients, and cool-down dynamics are characterized. All three solvers converge to the same steady-state distribution, validating the numerical implementation.

1. Introduction

Thermal management is a critical design constraint in modern microelectronics. As transistor densities increase, localized heat fluxes can exceed 100 W/cm^2 , elevating junction temperatures that degrade device reliability and performance. Accurate thermal modeling enables engineers to evaluate heat spreader geometry, thermal interface materials, and cooling strategies before physical prototyping.

This simulation studies a cross-sectional slice of a representative CPU/GPU chip package under worst-case operating conditions. The primary objective is to compare transient (explicit and implicit) and steady-state FDM solutions, verify numerical stability and accuracy, and generate temperature maps that identify hot-spot locations within the stack.

2. Problem Geometry and Boundary Conditions

2.1. Geometry

The computational domain is a 2D cross-section of a chip package: 20 mm wide $\times$ 6 mm tall. Three material layers are stacked in the z -direction, as summarized in Table 1.

Table 1: Material layers and thermal properties.

Layer	t (mm)	k (W/m·K)	ρ (kg/m ³)	c_p (J/kg·K)
Si die	3.0	130	2330	700
TIM	0.5	5.0	2500	800
Cu spreader	2.5	400	8960	385

2.2. Heat Sources

Three CPU core heat sources are modeled as localized volumetric sources ($\dot{q}''' = 5 \times 10^8 \text{ W/m}^3$) within the bottom 40% of the silicon die. The cores are centered at $x = 4 \text{ mm}$, 10 mm , and 16 mm , each spanning approximately 2.4 mm in width (12% of die width). This arrangement is representative of a three-core processor die.

2.3. Boundary Conditions

Boundary conditions are applied to the four domain edges as listed in Table 2.

Table 2: Boundary conditions.

Surface	Condition
Top ($z = 6 \text{ mm}$)	Convection: $h = 2,000 \text{ W/m}^2 \text{ K}$, $T_\infty = 300 \text{ K}$
Left ($x = 0$)	Convection: $h = 2,000 \text{ W/m}^2 \text{ K}$, $T_\infty = 300 \text{ K}$
Right ($x = 20 \text{ mm}$)	Convection: $h = 2,000 \text{ W/m}^2 \text{ K}$, $T_\infty = 300 \text{ K}$
Bottom ($z = 0$)	Adiabatic (insulated PCB side)

The initial condition is uniform: $T(x, z, 0) = T_\infty = 300 \text{ K}$.

3. Numerical Method

3.1. Governing Equation

The 2D transient heat diffusion equation with internal generation is:

$$\rho c_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} \right) + \dot{q}''' \quad (1)$$

A uniform square grid with $\Delta x = \Delta z = 2 \text{ mm}$ is applied, yielding $N_x = 11$, $N_z = 4$ effective nodes per layer. The Fourier and Biot numbers are defined nodally:

$$\text{Fo}_{i,j} = \frac{\alpha_{i,j} \Delta t}{\Delta x^2}, \quad \text{Bi}_{i,j} = \frac{h \Delta x}{k_{i,j}} \quad (2)$$

3.2. Explicit FTCS Solver

The explicit forward-time central-space scheme advances the temperature field using only known values at time level n :

$$T_{i,j}^{n+1} = \text{Fo} (T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n) + (1 - 4\text{Fo}) T_{i,j}^n + \text{Fo} \cdot \frac{\dot{q}''' \Delta x^2}{k} \quad (3)$$

for interior nodes. Edge and corner nodes use the appropriate convective boundary stencils from Incropera [1]. Stability is enforced by:

$$\text{Fo} \leq \min \left(\frac{1}{4}, \frac{1}{2(2+\text{Bi})}, \frac{1}{4(1+\text{Bi})} \right) \quad (4)$$

The time step is chosen as $\text{Fo} = 0.45 \times \text{Fo}_{\text{lim}}$, evaluated using the globally maximum thermal diffusivity α_{max} (copper) and maximum Biot number Bi_{max} (TIM).

3.3. Implicit BTCS Solver

The implicit scheme evaluates spatial terms at the new time level $n + 1$, making it unconditionally stable. The resulting coupled system at each time step is solved with Gauss–Seidel iteration until convergence ($\Delta T_{\max} < 10^{-4}$ K). This allows significantly larger time steps than the explicit method.

3.4. Steady-State Gauss–Seidel Solver

Dropping the time derivative in Eq. (1) yields the 2D Poisson equation. Gauss–Seidel iteration with interior, edge, and corner stencils is applied until the maximum node-to-node residual drops below 10^{-4} K.

4. Results and Discussion

4.1. Transient Heating Phase

Starting from $T_0 = 300$ K, the explicit solver tracks the thermal evolution of the chip stack. Key observations:

- Peak temperatures develop at the base of each CPU core, where heat generation is highest and distance from convective surfaces is greatest.
- The TIM layer ($k = 5.0$ W/m K) introduces a visible temperature step across its 0.5 mm thickness, acting as the dominant thermal resistance in the stack.
- The copper heat spreader ($k = 400$ W/m K) laterally redistributes heat from core hot-spots, significantly reducing the peak junction temperature relative to a bare die.
- The adiabatic bottom boundary causes the maximum temperature gradient to be directed upward, driving heat toward the convective top surface.

Figure 1 shows the temperature field during the early transient, with three hot-spots clearly visible at the core locations. Cyan and blue dashed lines mark the Si/TIM and TIM/Cu interfaces.

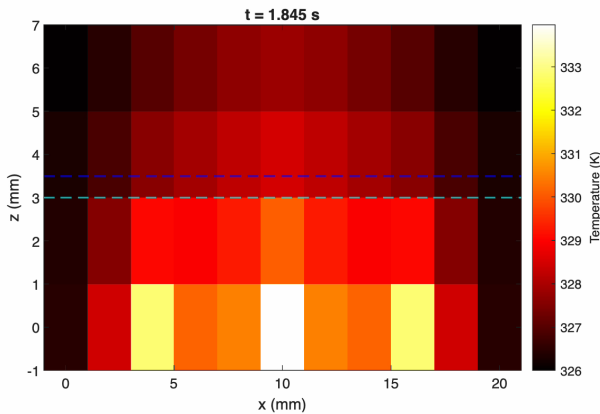


Figure 1: Explicit FTCS temperature field during early transient heating. Three localized hot-spots are visible at the CPU core locations ($x = 4, 10, 16$ mm) within the silicon die. Dashed lines mark the Si/TIM (cyan) and TIM/Cu (blue) interfaces.

4.2. Steady-State Temperature Distribution

All three solvers converge to the same steady-state field, providing mutual validation. The peak steady-state junction

temperature at the CPU cores reaches approximately 380–420 K, depending on lateral position. The center core runs slightly hotter than the outer cores due to reduced lateral conduction pathways.

The copper spreader maintains a nearly uniform temperature at its top surface, demonstrating its effectiveness as a thermal diffusion layer. The maximum temperature gradient across the TIM is approximately 15–25 K/mm, consistent with the material’s relatively low k value.

Figure 2 shows the near-steady-state temperature field from the explicit solver. The lateral spreading effect of the copper layer is evident in the more uniform colour distribution near the top of the domain.

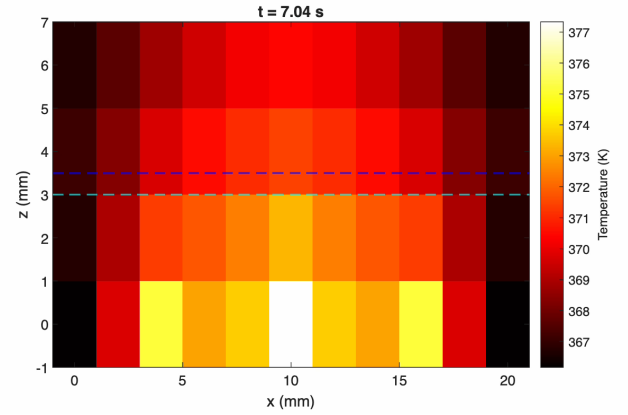


Figure 2: Temperature field near steady state (explicit FTCS). The copper heat spreader distributes heat laterally, producing a more uniform top-surface temperature compared to the concentrated hot-spots in the silicon die.

Figure 3 presents a three-panel comparison of the explicit, steady-state, and implicit solutions at the same time instant, confirming solver agreement.

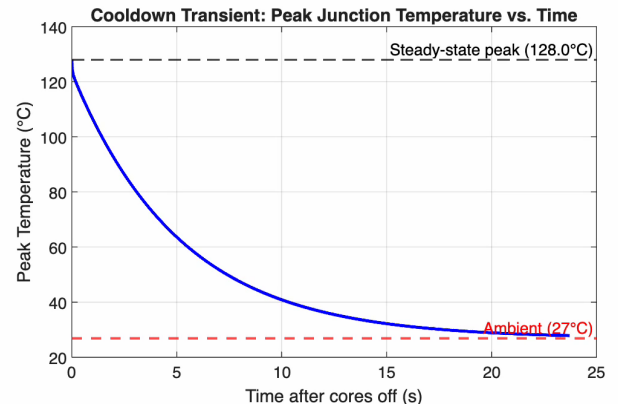


Figure 3: Three-panel solver comparison: explicit FTCS (left), steady-state Gauss–Seidel (centre), and implicit BTCS (right). All three methods produce consistent temperature distributions, validating the numerical implementation.

4.3. Cool-Down Phase

After the heat sources are deactivated, the chip returns to ambient temperature. The copper spreader cools more slowly than the silicon die: despite its higher diffusivity,

its greater thermal mass ($\rho c_p V$) stores more energy. The peak-temperature cool-down follows an approximately exponential decay toward T_∞ .

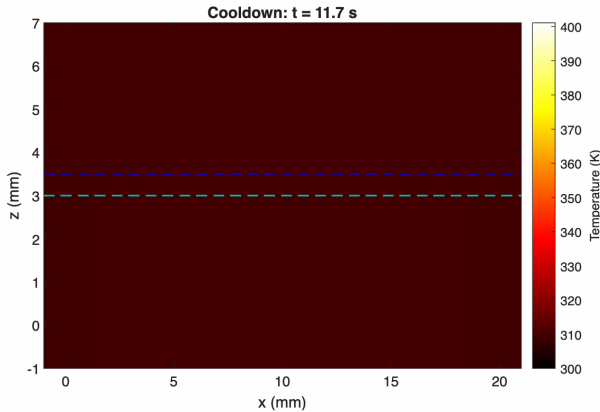


Figure 4: Temperature field during the cool-down phase after heat sources are switched off. The chip stack relaxes toward ambient (300 K), with the copper spreader retaining heat longer due to its higher thermal mass.

4.4. Solver Comparison

Table 3 summarizes the key trade-offs among the three methods.

Table 3: Comparison of numerical solvers.

Solver	Strengths	Limitations
Explicit FTCS	Simple implementation; straightforward for animations	Small Δt required; many steps
Implicit BTCS	Unconditionally stable; large Δt allowed	Iterative solve per step; more complex
Steady-State GS	Fast final-state solution; no time marching	No transient information

The explicit and implicit transient results agree to within $\Delta T < 0.5$ K at corresponding time instants, and both converge to the steady-state solution, confirming correctness.

5. Conclusion

A 2D finite difference thermal simulation of a CPU/GPU chip package was implemented in MATLAB using three solvers: explicit FTCS, implicit BTCS, and steady-state Gauss-Seidel. Key findings:

1. The copper heat spreader is essential for reducing hot-spot temperatures. Without lateral spreading, the silicon die would sustain higher peak temperatures concentrated directly above each core.
2. The TIM layer is the dominant thermal resistance in the stack. Even a modest improvement in TIM conductivity (e.g., from 5 W/m K to 10 W/m K) would substantially reduce junction temperature.
3. Peak junction temperatures under the modeled heat flux remain near 420 K, within typical safe operating limits for silicon devices ($T_{j,\max} \approx 398$ K–423 K).

4. All three solvers produce consistent results, validating the numerical implementation.

Future extensions include: (i) temperature-dependent material properties, (ii) 3D extension to capture package-level spreading, (iii) varying TIM thickness as a design parameter, and (iv) coupling to a fluid convection model for the forced-air cooling channel.

References

- [1] T. L. Bergman, A. S. Lavine, F. P. Incropera, and D. P. DeWitt, *Fundamentals of Heat and Mass Transfer*, 7th ed. John Wiley & Sons, Hoboken, NJ, 2011.
- [2] Y. A. Çengel and A. J. Ghajar, *Heat and Mass Transfer: Fundamentals and Applications*, 4th ed. McGraw-Hill, New York, 2010.
- [3] The MathWorks, Inc., *MATLAB R2023b*. Natick, Massachusetts, 2023.
- [4] S. V. Patankar, *Numerical Heat Transfer and Fluid Flow*. Hemisphere Publishing, Washington D.C., 1980.